

KNIME Analytics Platform

File Edit View Node Help

100%

KNIME Explorer

- My-KNIME-Hub (api.hub.knime.com)
- EXAMPLES (knime@api.hub.knime.com)
- LOCAL (Local Workspace)
 - Example Workflows
 - Demo1
 - 03_Sentiment_Classification.knwf
 - CutomerChurnData.csv
 - diabetes.csv
 - IMDb-sample.csv
 - iris.csv
 - marketbasket.arff
 - marketbasket.csv
 - MPQA-OpinionCorpus-NegativeList.csv
 - MPQA-OpinionCorpus-PositiveList.csv

Workflow Coach

Recommended Nodes Community

Node Repository

Views

- JavaScript
- ROC Curve

Analytics

- Mining
- Scoring
- ROC Curve (local)

Welcome to KNIME Analytics Platform

3: Demo1

CSV Reader Node 1

Color Manager Node 5

Partitioning Node 2

Decision Tree Learner Node 3

Decision Tree Predictor Node 4

ROC Curve Node 6

Outline

Console

Node Monitor

KNIME Console

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WARN Decision Tree Predictor 3:4 DataColumnSpec already con
WARN Decision Tree Predictor 3:4 DataColumnSpec already con
WARN Decision Tree Predictor 3:4 DataColumnSpec already con
WARN Decision Tree Predictor 3:4 DataColumnSpec already con
WARN ROC Curve 3:6 Class column 'null' does not
```

3:12 PM 8/30/2025

Activity Evidence

Rewatch 'Getting Started' if needed.

Creating a Decision Tree in KNIME

Please remember

- After completing the task, click **Submit** and select **Mark Attempted** to indicate that you have attempted the activity.
- Please take any additional steps prescribed by your school.
- If any software update prompt appears, please close or dismiss it.

In this lab, you will learn to create a decision tree in KNIME. Decision trees can be used to predict both continuous and discrete values, that is, they work well for both regression and classification tasks.

To create a decision tree in KNIME, here's what you need to do:

- From the desktop, double-click **KNIME Analytics Platform**.
- In the **KNIME Analytics Platform Launcher** dialog box, ensure that **C:\LabFiles** is selected as the workspace. If not, then click **Browse** and perform the following steps:
 - In the **Select Workspace Directory** window, in the left pane, click **This PC**.
 - In the right pane, scroll down to double-click **Local Disk (C:)**, and then click **LabFiles > Select Folder**.

- In the **KNIME Analytics Platform Launcher** dialog box, click **Launch**.

Note: If the **Startup of KNIME Analytics Platform** is taking long dialog box appears, click **OK**. Also, if the **Help Improve KNIME** dialog box appears, click **No**.

- Create a workflow and read the dataset.

- In the left pane, under **KNIME Explorer**, right-click **LOCAL (Local Workspace)**, and select **New KNIME Workflow**.
- In the **New KNIME Workflow Wizard**, type **Name of the workflow to create as Demo1**, and then click **Finish**.
- In the **KNIME Explorer** pane, scroll down and drag **iris.csv** and drop it to the **Demo1** workflow.
- In the **Dialog - 3-1 - CSV Reader** dialog box, analyze the data and click **OK**.
- In the upper-middle pane, right-click **CSV Reader** and select **Execute**.

- In the left pane, under **Node Repository**, in the search box, type **partition**, click **Partitioning**, and then drag and drop it to the **Demo1** workflow.
- In the **Demo1** workflow, drag the right arrow of **CSV Reader (Node 1)** and drop it to the left arrow of **Partitioning (Node 2)** to connect to it.
- Right-click **Partitioning** and select **Configure**.
- In **Dialog - 3-2 - Partitioning**, select **Stratified sampling**, and then click **Apply > OK**.
- Right-click **Partitioning** and select **Execute**.
- Hover on the first right arrow of **Partitioning** to see the number of rows and then on the second right arrow to see the number of rows.

- Use a decision tree algorithm to train and test the data.

- In the left pane, under **Node Repository**, in the search box, type **decision**, click **Decision Tree Learner**, and then drag and drop it to the **Demo1** workflow.
- From **Node Repository**, click **Decision Tree Predictor**, and then drag and drop it to the **Demo1** workflow.
- In the **Demo1** workflow, drag the first right arrow of **Partitioning (Node 2)** and drop it to the left arrow of **Decision Tree Learner (Node 3)** to connect to it.
- Right-click **Decision Tree Learner** and select **Configure**.
- In **Dialog - 3-3 - Decision Tree Learner**, scroll-down and check the **Binary nominal splits** checkbox, and then click **Apply > OK**.
- Right-click **Decision Tree Learner** and select **Execute and Open Views**.
- In the **Decision Tree View - 3-3 - Decision Tree Learner** window, analyze the decision tree, and then close the window.
- In the left pane, under **Node Repository**, in the search box, type **color**, click **Color Manager**, and then drag and drop it in between **CSV Reader** and **Partitioning** on the **Demo1** workflow.
- In the **Confirm** dialog box, click **OK**.
- In the **Confirm reset** dialog box, click **OK**.
- Right-click **Color Manager** and select **Configure**.
- In **Dialog - 3-5 - Color Manager**, click **Apply > OK**.
- Right-click **Decision Tree Learner** and select **Execute and Open Views**.
- In **Decision Tree View - 3-3 - Decision Tree Learner**, analyze the decision tree, and then close the window.
- On the **Demo1** tab, drag the blue-colored port of **Decision Tree Learner** and drop it to the blue-colored port of **Decision Tree Predictor** to connect to it.
- Drag the second right arrow of **Partitioning (Node 2)** and drop it to the left arrow of **Decision Tree Predictor (Node 4)** to connect to it.
- Right-click **Decision Tree Predictor** and select **Configure**.
- In **Dialog - 3-4 - Decision Tree Predictor**, check the **Append columns with normalized class distribution** checkbox, type **Suffix for probability columns as _PT**, and then click **Apply > OK**.
- Right-click **Decision Tree Predictor** and select **Execute**.
- Right-click **Decision Tree Predictor**, select **Classified Data** to observe the data, and then close the **Classified Data - 3-4 - Decision Tree Predictor** window.

- Create an ROC curve to identify the accuracy of prediction.

- In the left pane, under **Node Repository**, in the search box, type **ROC**, click **ROC Curve**, and then drag and drop it to the **Demo1** workflow.
 - In the **Demo1** workflow, drag the right arrow of **Decision Tree Predictor** and drop it to the first left arrow of **ROC Curve** to connect to it.
 - Right-click **ROC Curve** and select **Configure**.
 - In **Dialog - 3-6 - ROC Curve**, from the **Class** column drop-down list, select **variety**, click the **Moves all visible columns from the right to the left list** (**<>**) button to move every columns to the left list, select **P (variety=Setosa)**, **DT** from the left list, click the **Move the selected columns from the left to the right list** (**><**) button to move the column to the right list, and then click **Apply > OK**.
 - Right-click **ROC Curve**, select **Execute and Open Views**, observe the curve, and then close the **ROC Curve** window.
- Note:** Wait for 2-3 seconds if the **ROC Curve** is not visible.
- Again, right-click **ROC Curve** and then select **Configure**.
 - In **Dialog - 3-6 - ROC Curve**, from the **Positive class value** drop-down list, select **Virginica**, click the **Moves all visible columns from the right to the left list** (**<>**) button to move column to the left list, select **P (variety=Virginica)**, **DT** from the left list, click the **Move the selected columns from the left to the right list** (**><**) button to move the column to the right list, and then click **Apply**.
- Note:** In the **Confirm reset** dialog box, click **OK**.
- In **Dialog - 3-6 - ROC Curve**, click **OK**.
 - Right-click **ROC Curve**, select **Execute and Open Views**, and then observe the curve.

- Take the screenshot and upload your work. ([Click here](#) to follow the instructions.)

- Close the **ROC Curve** window.

- Close the **Demo1** workflow.

- In the **Save Resource** dialog box, click **Yes**.

- Close all windows.



Lesson

Introduction to Predictive Analytics and Data Mining

Popular Data Mining Tools



Win -



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◀ PREVIOUS

2 of 2

NEXT ▶



SUBMIT



CLOSE